

Oct 22

Consider  $\sum_{k=2}^{\infty} a_k$

$$a_k = \frac{k+2}{k^3+2}$$

$$a_k = \frac{k+2}{k^3+2} \leq \frac{k+2}{k^3}$$

$$b_k = \frac{2}{k^2} + \frac{2}{k^3}$$

$$b_k \geq a_k$$

b/c denom of  $b_k$  is smaller + that is only diff

bigger:

$$\begin{aligned} & \frac{k+2}{k^3} \\ &= \frac{k}{k^3} + \frac{2}{k^3} \\ &= \frac{1}{k^2} + \frac{2}{k^3} \\ &= \frac{1}{k^2} + \frac{2}{k^3} \end{aligned}$$

$\frac{1}{k^2}$  p-series,  $p=2 > 1$  conv.  
 $\frac{2}{k^3}$  p-series,  $p=3 > 1$  conv.

can be broken into pieces if each

indiv. piece converges

Both are p-series when

$p > 1$  so they both

converge so their

sum will also converge

$b_k$  conv. b/c  $b_k$  is sum of 2 conv. p-series ( $p=3$  and  $p=2$ ) so since  $b_k \geq a_k$ , by BCT  $a_k$  converges

Using LCT

$$c_k = \frac{1}{k^2}$$

$$\lim_{k \rightarrow \infty} \frac{a_k}{c_k} = \lim_{k \rightarrow \infty} \frac{k+2}{k^3+2}$$

$$\left[ \frac{k+2}{k^3+2} \right]_{\infty}^{\infty}$$

$$\text{LH} \rightarrow \frac{3k^2+2k}{3k^2} \left[ \right]_{\infty}^{\infty}$$

$$\text{LH} \rightarrow \frac{6k+2}{6k} \left[ \right]_{\infty}^{\infty}$$

$$\text{LH} \rightarrow \frac{6}{6} \rightarrow \lim_{k \rightarrow \infty} \frac{a_k}{c_k} = \frac{6}{6} = 1 \neq 0 \neq \infty$$

both converge

You could also

see... highest powers are same

$$\lim_{k \rightarrow \infty} \frac{a_k}{c_k} = \frac{2}{2} = 1$$

so limit proves both conv